Assignment 2

Econ 8007 Macroeconomic Theory I, Fall 2026

Tuesday, September 23, 2026

Throughout, use the baseline monthly calibration for the business cycle model with labor market frictions:

Parameter	Value	Description
β	0.996	discount factor (monthly; implies $r \approx 0.004$)
σ	0.02	separation rate
η	0.5	elasticity of matching w.r.t. unemployment
γ	0.5	worker's bargaining weight
z	1	productivity (baseline level)
b	0.4	flow value of unemployment
χ	0.53	matching efficiency
κ	0.75	flow cost of a vacancy

Part A. Steady state

A1. Solve the equilibrium condition

$$\frac{(r+\sigma)\,\kappa}{\lambda_f(\theta)} + \gamma\kappa\theta = (1-\gamma)(z-b)$$

for θ^* (you should solve it numerically). Report θ^*, λ_w, λ_f, w, u^*, the joint surplus S, and the firm's value J. Verify the free-entry identity $\kappa/\lambda_f(\theta^*) = \beta(1-\gamma)S$ also numerically.

A2. Plot u^* as a function of $b \in [0, 0.95]$. Discuss that result.

Part B. The Shimer exercise in steady-state

B1. Set $\gamma = 0.05$ and find the value of b that leaves the steady state (θ^*, u^*) exactly unchanged. Report that b and the new $\varepsilon_{\theta z}$.

B2. Plot $\varepsilon_{\theta z}$ against $b \in [0, 0.96]$ for $\gamma \in \{0.5, 0.05\}$, marking the baseline and the point from B1. Indicate the data-consistent level $\varepsilon \approx 19$ on the figure.

B3. At both calibrations (baseline and B1), compute $du/d\ln b$ numerically. Then answer: if empirical studies bound the effect of a 10% increase in benefit generosity on the unemployment rate at 0.3 percentage points, which calibration survives? Write one paragraph on why the volatility fix and the policy prediction are the same elasticity.

Part C. The fully stochastic case

Let z_t (the value of productivity) follow a five-state Markov chain with the following transition matrix

$$P = \begin{pmatrix} 0.960596 & 0.038812 & 0.000588 & 0.000004 & 0.000000 \\ 0.009703 & 0.960890 & 0.029112 & 0.000294 & 0.000001 \\ 0.000098 & 0.019408 & 0.960988 & 0.019408 & 0.000098 \\ 0.000001 & 0.000294 & 0.029112 & 0.960890 & 0.009703 \\ 0.000000 & 0.000004 & 0.000588 & 0.038812 & 0.960596 \end{pmatrix}$$

and 5 states:

$$z_i \in \{0.93771,\ 0.96835,\ 1,\ 1.03268,\ 1.06643\}\,.$$

(This is the $N = 5$ Rouwenhorst discretization of the monthly AR(1) process $\ln z_{t+1} = 0.98\,\ln z_t + \varepsilon_{t+1}$ with $\varepsilon_{t+1} \sim N(0,\, 0.0064^2)$; the implied unconditional standard deviation of $\ln z$ is 0.0322.)

Throughout this part, $\lambda_w(\theta) = \chi\,\theta^{1-\eta}$ and $\lambda_f(\theta) = \chi\,\theta^{-\eta}$, and along any simulated path $\{z_t\}$ unemployment evolves according to

$$u_{t+1} = u_t + \sigma\,(1 - u_t) - \lambda_w(\theta(z_t))\,u_t, \qquad v_t = \theta(z_t)\,u_t.$$

In the simulations, cap λ_w at one whenever the formula above exceeds it.

C1. Generate a long time series of z_t (e.g. 50,000 periods). Discard the first few thousands (e.g. first 5,000).

C2. *Nash Wages.* Solve the pair $\{S(z_i), \theta(z_i)\}_{i=1}^{5}$ on the grid by iterating jointly on the surplus Bellman equation and the free-entry condition:

$$S(z_i) = z_i - b + \beta\,\big[\,1 - \sigma - \gamma\,\lambda_w(\theta(z_i))\,\big]\sum_{j=1}^{5} P_{ij}\,S(z_j),$$

$$\frac{\kappa}{\lambda_f(\theta(z_i))} = \beta\,(1-\gamma)\sum_{j=1}^{5} P_{ij}\,S(z_j).$$

Concretely: given a current guess $S^{(k)}(\cdot)$, invert the free-entry condition state by state to obtain $\theta^{(k)}(z_i)$, substitute it into the Bellman equation to compute the update $S^{(k+1)}(\cdot)$, and repeat until $\max_i \big|S^{(k+1)}(z_i) - S^{(k)}(z_i)\big|$ falls below your tolerance. (Sanity check: with P replaced by the identity matrix and $z_i = 1$, this system collapses to the steady-state conditions of A1.) Note that the bargained wage never appears in this system: the Nash

sharing rule is already embedded in the surplus recursion. Once you have $\theta(z_i)$, recover the wage state by state from

$$w(z_i) = \gamma\, z_i + (1-\gamma)\, b + \gamma\, \kappa\, \theta(z_i).$$

Simulate, and report the moment table: sd of $\ln z, \ln \theta, \ln u, \ln w$; the ratios $\mathrm{sd}(\ln\theta)/\mathrm{sd}(\ln z)$ and $\mathrm{sd}(\ln w)/\mathrm{sd}(\ln z)$; and $\mathrm{corr}(\ln u, \ln v)$. Compare the ratio with the baseline steady-state $\varepsilon_{\theta z}$ from B2 (the $\gamma = 0.5$, $b = 0.4$ point) and the correlation with the data's -0.89.

C3. *Fixed wages.* Fix the wage at the steady-state Nash wage from A1,

$$\bar{w} = \gamma\left(\bar{z} + \kappa\,\theta^*\right) + (1-\gamma)\, b, \qquad \bar{z} = 1,$$

and re-solve. Only the firm's value is now needed. Because the wage no longer adjusts, low realizations of z can drive the value of the match to the firm below the value of walking away; impose the firm's participation floor directly in the Bellman equation and iterate on

$$J(z_i) = \max\{\, 0,\ \tilde{J}(z_i)\,\}, \qquad \tilde{J}(z_i) \equiv z_i - \bar{w} + \beta\,(1-\sigma)\sum_{j=1}^{5} P_{ij}\, J(z_j),$$

where $\tilde{J}(z_i)$ is the *pre-floor* firm value. (Without the max operator this would be a linear system you could solve in one step; with it, iterate to convergence.) Market tightness then follows state by state from free entry,

$$\frac{\kappa}{\lambda_f(\theta(z_i))} = \beta \sum_{j=1}^{5} P_{ij}\, J(z_j) \quad \Longrightarrow \quad \theta(z_i) = \left(\frac{\chi\,\beta}{\kappa}\sum_{j=1}^{5} P_{ij}\, J(z_j)\right)^{1/\eta}.$$

Before simulating, predict the volatility ratio from the rigid-wage (fundamental-surplus) formula in the Topic 3 notes, evaluated at the steady state:

$$\frac{\mathrm{sd}(\ln\theta)}{\mathrm{sd}(\ln z)} \;\approx\; \varepsilon_{\theta z}^{\text{rigid}} \;=\; \frac{1}{\eta}\cdot\frac{\bar{z}}{\bar{z}-\bar{w}},$$

where $\mathrm{Y} = \bar{z} - \bar{w}$ is the fundamental surplus of the rigid-wage regime. Then simulate. You will find the simulated ratio lands in the same ballpark as the formula's prediction—but this agreement conceals a violently asymmetric response, with $\ln\theta$ collapsing in the lowest-z state and moving by much less than the formula predicts in high-z states. Explain the asymmetry quantitatively: report the fraction of months in which the pre-floor firm value is negative, $\tilde{J}(z_t) < 0$; connect this to the width of the bargaining set (Topic 3, Exercise 27); and explain why $\ln\theta$ is so sensitive precisely in low-z states. What does this experiment teach about the domain of validity of a fully rigid wage norm at a thin profit margin?